

Physics Internal Assessment

How does changing wind speeds affect lift force generated?

Introduction:

Since a young age, it has always been my dream to fly. When I finally got the opportunity to fly at age 17, the experience was an exhilarating one. During the lessons at flight school, we got taught the basics and foundations for aerodynamics and this sparked an interest in me for the mechanics of flight. During the course of flight school, we would learn about take-off speeds and certain speeds that were needed to be taken note of. These speeds vary from aircraft to aircraft, based on factors like wingspan or weight and the greater aircraft themselves have their own speed restrictions to keep within. With little information given about calculating the speeds and weights during training itself, I began to explore how these weights were calculated, and that is where I came across the lift equation. Upon some reading, I realized that it may be possible to carry out an experiment investigating the effects of windspeeds on lift force in my own school laboratory. Therefore, I created a scaled down model of a wing, and carried out my own experiment.

Research Question: How does the wind speed affect the amount of lift generated by an aerofoil?

Background:

Bernoulli's Principle:

In his book, *Hydrodynamica* (1738), the Bernoulli talks about what is now named after him, the Bernoulli's Principle. In fluid dynamics, we learn about Bernoulli's principle, which states that an increase in the speed of a fluid occurs simultaneously with a decrease in pressure or a decrease in the fluid's potential energy. (SKYbrary, Bernoulli's Principle, 2022) This can be represented in the equation below, taken from "Higher Level Physics – Chris Hamper second edition":

$$P + \frac{1}{2} \rho v^2 = \text{constant}$$

Where P refers to the local hydrostatic pressure, ρ represents the density, and v refers to the velocity of the fluid. This equation can then be implemented in the figure shown below, which demonstrates how Bernoulli's principle can be implemented in a tube of changing diameters.

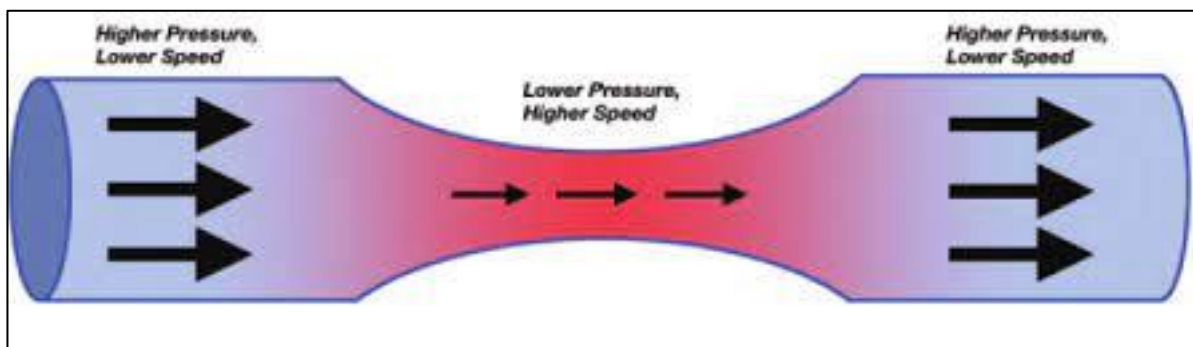


Fig. 1 (Visual representation of relationship between pressure and speed, adapted from (ResearchGate, 2021))

This shows that when compressed into a smaller area, as pressure decreases, the speed of the air will increase. The reverse could also happen, as if the speed of airflow increases, the pressure of the air at that place will decrease. The law of conservation of energy states that energy can be neither destroyed nor created, but simply converted from one form to another, hence the total energy within an isolated system remains the same. This can be represented in the equation below, which combines the equation previously shown and the law of conservation of energy. (The Trustees of Princeton University. (n.d.))

$$P_1 + \frac{1}{2} \rho (v_1)^2 = P_2 + \frac{1}{2} \rho (v_2)^2$$

Coanda Effect:

The Coanda effect is a phenomenon in which a stream of air or gas attaches itself to a nearby surface, and follows that surface even if it curves away from the original trajectory of the jet stream. (S&C Thermofluids Ltd, 2014)

This occurs because when the air reaches the curved surface, it pushes away the air that was already there, and this creates an area of low pressure, which the incoming air proceeds to flow through. As a result, the air surrounding this has a higher pressure, and forces the stream of air into the curved surface, not allowing it to deflect off the surface. This can be illustrated in the figure shown on the right.

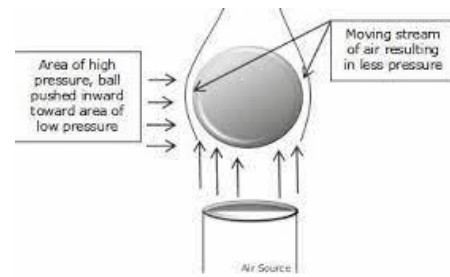


Fig. 2 (Visual representation of Coanda Effect on Ping Pong ball, adapted from (*Bernoulli's: Demos*, n.d.))

About aerofoils:

Aerofoils are the cross section of the wing of an aircraft, which help it to generate lift in and get the aircraft off the ground. This works through the combination of the concepts of Bernoulli's Principle and the Coanda effect. When the aircraft accelerates, the airflow around the wing of the aircraft increases, and due to the Coanda effect, the airflow gets directed over the curved surface on the top of the wing. As this happens, the speed of the airflow over the top of the wing increases, and by law of conservation of energy in Bernoulli's Principle, the pressure over the top of the wing hence decreases. (*Principles of flight: Foam wing (grades K-12)*, n.d.). However, since the bottom of the wing is not as curved and mostly flat, the speed of airflow does not increase as much, and this results in the pressure below the wing being higher than the pressure above the wing. As the airflow increases in speed, the difference in pressure also increases, which results in an upward force being exerted on the wing of the aircraft. This force is known as the Lift Force. This can be demonstrated in the diagram below:

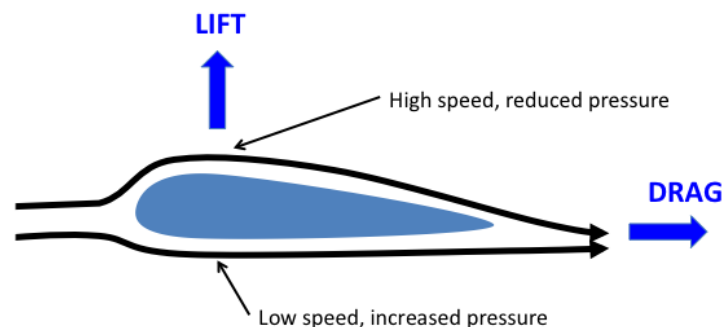


Fig. 3 (Diagram of Aerofoil showing Coanda effect, adapted from SKYbrary Aviation Safety. (2021, September 28))

So how is Lift calculated?

Lift depends on the various factors, like the density of air, the surface area of the wing and the velocity of the incoming airflow squared. There are other factors, like the air's viscosity and compressibility, and the inclination of the body to the flow of the air. However, in most cases, like the experiment being carried out these factors remain constant, and hence these factors can be characterized into a single variable, known as the coefficient of lift, or "Cl". With this knowledge, the lift equation can be shown below, where L is the lift, v is the velocity of the incoming air, A is the surface area of the top and the bottom of the wing combined, and ρ is the density of air. (The Lift Coefficient, NASA, 2021)

$$L = Cl * \frac{\rho * v^2}{2} * A$$

Experimental Design

Hypothesis

The lift force generated will increase as the windspeed increases. The lift force however, will increase proportionally the square of the windspeed. This can be represented in the equation shown below:

$$L = \left(Cl * \frac{\rho}{2} * A \right) * v^2$$

In this equation, since Cl , ρ , and A stay constant throughout the experiment, this value can be expressed in a single constant k .

$$L = k * v^2$$

From the experiment, when the values of L are plotted against that of v^2 , the gradient of the line will be equal to k , or the product of Cl , ρ , and A divided by 2.

Variables

Independent variable – windspeed generated by the fan. Windspeeds of 2.0m/s, 2.5m/s, 3.0 m/s, 3.5m/s, 4.0 m/s and 4.5m/s. All the windspeeds were measured using a Kestrel 3000 Wind Meter, with an uncertainty of ± 0.1 m/s.

Dependent Variable – the difference in mass, which was measured using a weighing scale. The value was taken when the wind speed was stable and the weight remained constant for 2 or more seconds. The weighing scale had an uncertainty of ± 0.1 g.

Controlled Variables – the mass of the entire structure, measured at the beginning ($117.2\text{g} \pm 0.1\text{g}$); surface area of the wing ($589\text{cm}^2 \pm 1\text{cm}^2$); the density of air ($1005.6\text{ hPa} \pm 0.1\text{ hPa}$); distance from fan to wing ($60.0\text{cm} \pm 0.1\text{ cm}$); the height of the aerofoil and fan from the table ($20.0\text{cm} \pm 0.1\text{cm}$); the angle of attack of the airfoil (angle at which the aerofoil is in relation to the table).

Apparatus List

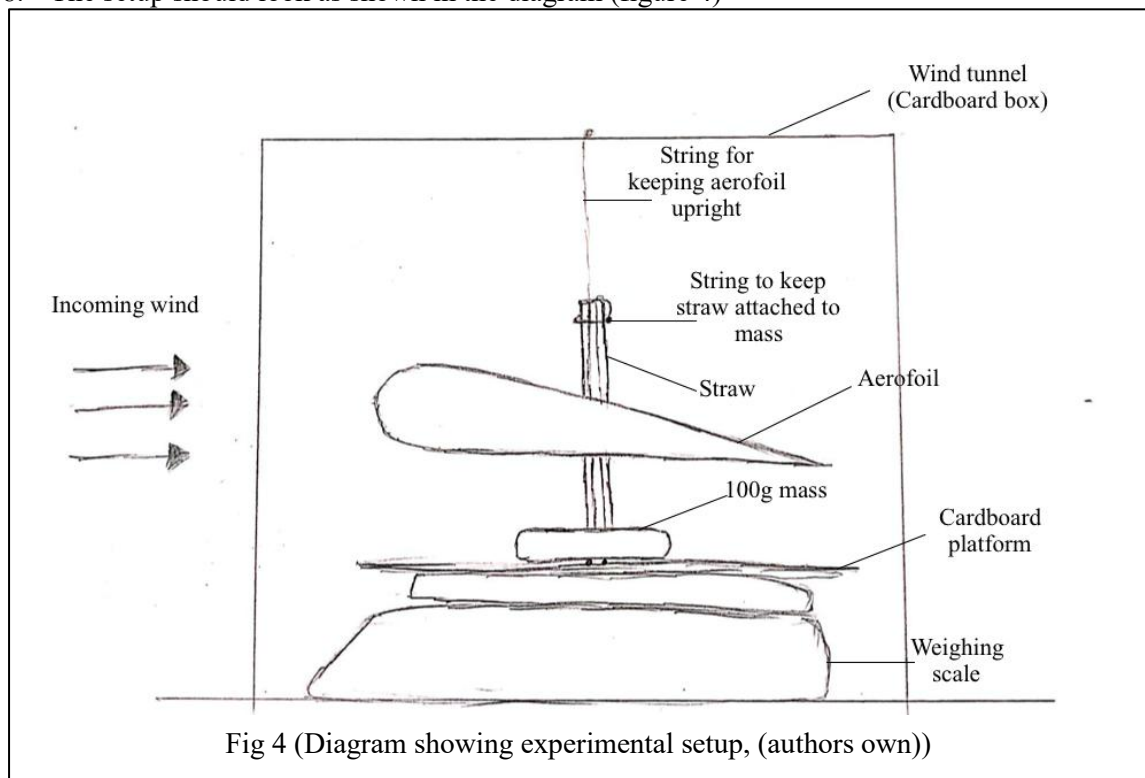
- 1 piece of A4 sized Cardstock
- 1 straw
- 1 fan of variable windspeeds
- 1 measuring scale
- 1 100g mass
- 1 Cardboard box
- 1 Ball of string
- 1 Piece of Cardboard
- 1 Penknife (handle with care)
- 1 Roll of tape
- 1 Kestrel 3000 Windspeed Meter

Assumptions

- The windspeeds did not change after the kestrel 3000 meter had been taken away.
- The only factor increasing the lift force of the aerofoil was the fan.
- All of the wind that was directed at the aerofoil was converted into lift, and none of it was converted into drag.
- The stabilized weight is the correct weight for that windspeed.

Procedure

1. Taking a string, pass it through the straw and tie it to the 100g weight, such that the weight and straw cannot move independent of each other.
2. Taking a piece of cardstock paper, fold the paper and measure tape the sides together, ensuring that the paper is not creased in the process. This creates the aerofoil.
3. Punch holes through the center of the aerofoil and pass the straw through it.
4. Taking a cardboard box, cut out one side such that it becomes the form of a tunnel.
5. Tape the weight to a cardboard platform and place it on a weighing scale in the wind tunnel.
6. Using another string, pass it through the straw again, and tape this to the cardboard base, with the other end of the string passing through a gap in the roof of the tunnel.
7. Ensure that the second string is always taught, to keep the aerofoil upright at all times.
8. The setup should look as shown in the diagram (figure 4)



9. Place the fan 60cm away from the aerofoil, and at the same level as the aerofoil.
10. Swing the kestrel 3000 wind meter clockwise 10 times and anticlockwise 10 times for calibration.
11. Take note of the initial weight shown on the weighing scale.
12. Turn the fan on, and place the wind meter 1cm from the front of the aerofoil to measure wind speed, and adjust the fan power to achieve a windspeed of 2.0m/s
13. Remove the wind meter, and take note of the reading on the weighing scale after the weight has stabilized.
14. Repeat steps 12 and 13 two more times for the same wind speed.
15. Repeat steps 12-14 for wind speeds of 2.5m/s, 3.0m/s, 3.5m/s, 4.0m/s and 4.5m/s.

Risk Assessment

The safety issues of this experiment are extremely minimal. The fan wind speeds were fairly fast, and so it was important that there were no light hazardous materials that were close to the fan. The handling of a scissors and penknife to cut out the aerofoil and cardboard box were also done safely using a mat and away from others to ensure no damage was done to anything else.

Results

Qualitative data

For all experiments, the wing was always kept attached to the weight and cardboard platform. It was noted that for higher windspeeds, the wing would occasionally shake a little, but this may have just been due to a slight inconsistency in the windspeed. However, since the numerical data was collected after the weight stabilized, this was not reflected in the quantitative data.

Raw data

Before carrying out the experiment, it was first important to measure the surface area of the wing. This was done by measuring three points along each cross section of the paper before it was folded into a wing. The three values were then averaged, and the length was multiplied by the breadth to find the surface area of the wing. The table below shows the data taken from this:

Table 1 (dimensions of the aerofoil)

Side	Length 1 (cm) (± 0.1)	Length 2 (cm) (± 0.1)	Length 3 (cm) (± 0.1)	Average Length (cm) (± 0.1)
Length	19.8	19.9	20.0	19.9
Breadth	29.5	29.6	29.6	29.6
			Surface Area (cm ²)	589.0 $\pm$ 5
			Surface Area (m ²)	0.00589

Firstly, it was important to find the base weight of the aerofoil. This was done by measuring it 3 times and finding the average, as demonstrated in the table below:

Table 2 (Base weight of the Experimental setup)

Trial Number	Base Weight (g) (± 0.1)
Trial 1	117.2
Trial 2	117.2
Trial 3	117.1
Average	117.2

For each wind speed, the new weight was measured after the weight stabilized, meaning that it stopped for more than 2 seconds, and the light appeared next to the weight indicating that the weight recorded was stable. This was repeated 3 times for each wind speed. The table on the next page shows the raw data collected from the experiment.

Table 3 (raw data collected from the experiment)

	New weight (g) ± 0.1		
Windspeeds (m/s) (± 0.1)	Trial 1	T2	T3
2.0	114.0	113.9	114.1
2.5	111.0	110.5	110.6
3.0	107.5	107.5	107.4
3.5	103.6	103.7	103.8
4.0	98.5	98.7	98.6
4.5	93.5	93.6	93.5

Calculations of uncertainties for mass and windspeeds:

For the windspeeds, the uncertainties were to the smallest decimal place shown on the windspeed mete, which was 0.1 m/s

For the mass, two methods of uncertainty had to be compared, in the forms of random error and systematic error.

For systematic error, the uncertainty was calculated in the method shown below:

Uncertainty of original weight = ± 0.1 (as on the weighing scale)

Uncertainty of new weight = ± 0.1

Uncertainty of weight difference = Uncertainty of original weight + Uncertainty of new weight

Uncertainty of weight difference = $(\pm 0.1 + \pm 0.1) = \pm 0.2$

For random error, the uncertainty was calculated in the method shown below, using the sample calculation of windspeed 2.0 m/s:

Uncertainty of weight difference = (maximum weight – minimum weight)/2

Uncertainty of weight difference = $(3.1 - 2.9)/2 = \pm 0.1$

In this case, since the systematic error was 0.2, and random error was 0.1, the systematic uncertainty was higher, and hence used over the random uncertainty in calculating the uncertainty of the experiment.

For the other windspeeds (2.5 m/s, 3.0 m/s, 3.5 m/s, 4.0 m/s and 4.5 m/s), these calculations were used as well, so as to help propagate the most accurate representation for the uncertainty of the changes in weight.

Processed data

Table 4 (Processed data)

	Weight difference (g) ± 0.2				
Windspeeds (m/s) (± 0.1)	T1	T2	T3	Average (g) (± 0.2)	Average (kg) (± 0.0002)
2.0	3.0	3.1	2.9	3.0	0.0030
2.5	6.5	6.4	6.6	6.5	0.0065
3.0	9.7	9.7	9.8	9.7	0.0097
3.5	13.6	13.5	13.4	13.5	0.0135
4.0	18.7	18.5	18.6	18.6	0.0186
4.5	23.7	23.6	23.7	23.7	0.0237

Sample calculation for windspeed of 2.0 m/s:

Weight difference (g) = $117.2 - 114.0 = 3.2$ (1 decimal place)

Uncertainty of original weight = ± 0.1

Uncertainty of new weight = ± 0.1

Uncertainty of weight difference = Uncertainty of original weight + Uncertainty of new weight

Uncertainty of weight difference = $(\pm 0.1 + \pm 0.1) = \pm 0.2$

Average weight difference (g) = (trial 1 + trial 2 + trial 3) / 3

Average weight difference (g) = $(3.2 + 3.1 + 2.8)/3 = 3.0$ (1 decimal place)

Uncertainty of average weight = $(\pm 0.2 + \pm 0.2 + \pm 0.2)/3 = \pm 0.2$

Table 5 (Table showing independent and dependant variables to be plotted on graph)

Windspeeds squared (m ² /s ²)	Uncertainty of Windspeeds Squared (m ² /s ²)	Lift force generated (N)	Uncertainty of Lift Force Generated (N)
4.00	± 0.4	0.030	± 0.002
6.25	± 0.5	0.064	± 0.002
9.00	± 0.6	0.095	± 0.002
12.25	± 0.7	0.132	± 0.002
16.00	± 0.8	0.182	± 0.002
20.25	± 0.9	0.232	± 0.002

Sample calculation for windspeed of 2.0 m/s:

With the knowledge that gravity has the literature value of 9.81N/kg, the amount of lift force can be calculated as shown below.

Lift force generated (N) = weight (kg) x 9.81N/kg

Lift force generated (N) = 0.0030 kg x 9.81N/kg = 0.029 N (2 significant figures)

Since 9.81 is a literature value, its uncertainty is negligible and the uncertainty of Lift can be calculated below:

Uncertainty of lift force = $(0.0002/0.0030 + 0/9.81) \times (0.029) = \pm 0.002$ (1 significant figure)

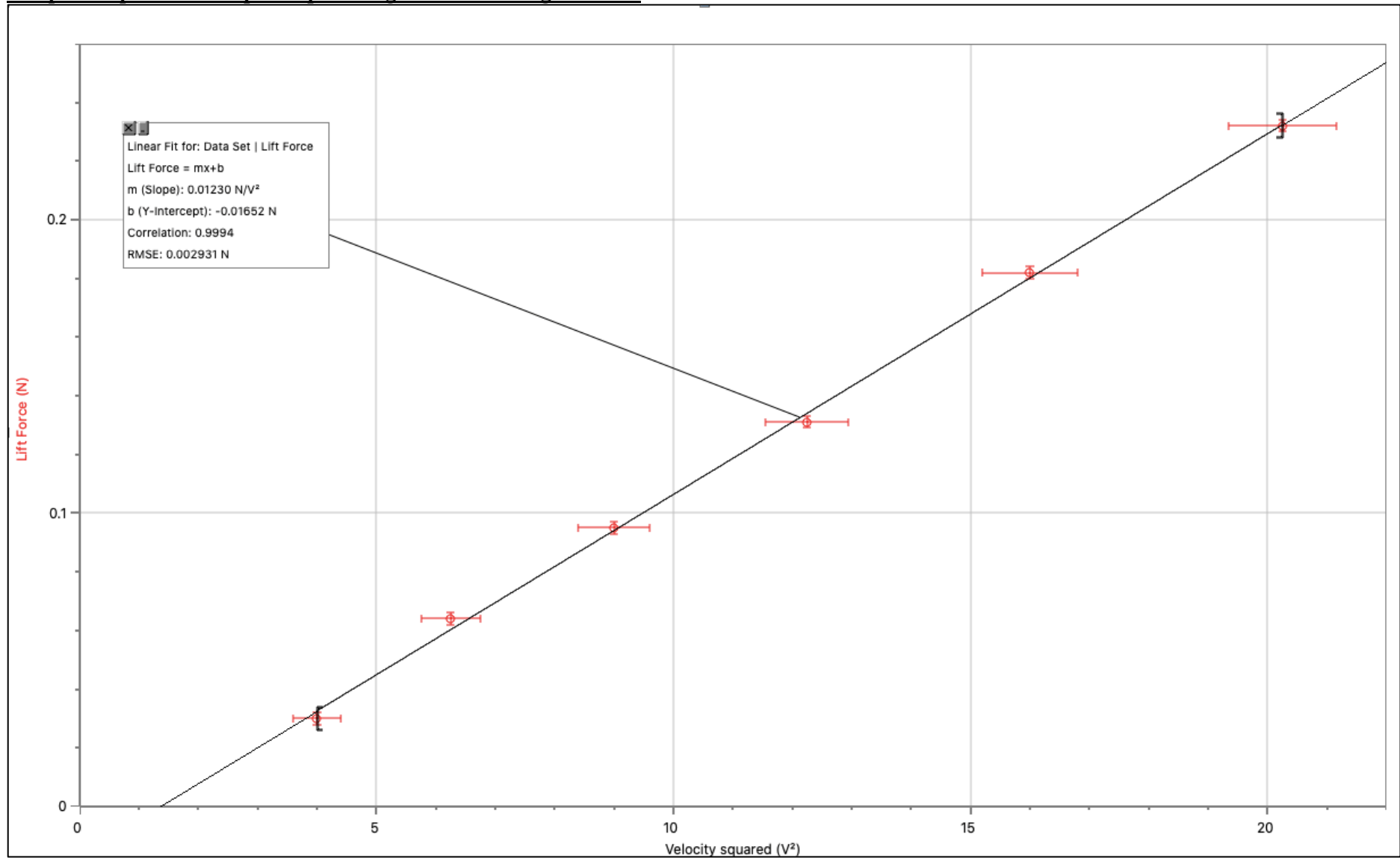
To calculate the value of the wind speed squared, the calculations are shown below

Windspeed squared (m²/s²) = original windspeed x original windspeed

Windspeed squared (m²/s²) = 2.0 x 2.0 = 4.0

Uncertainty of Windspeed squared = $(0.1/2.0 + 0.1/2.0) \times (4.0) = \pm 0.4$

Sample Graph for Windspeed squared against Lift force generated:



From this graph, a positive linear relationship can be observed between the lift force generated and the windspeed squared v^2 . The high correlation of 0.999 meant that the data points were almost all concordant with the line of best fit. Furthermore, all of the six error bars all touched the lines of best fit, and this indicated that the results were very reliable. However, the graph did not pass through the origin, indicating that the graph was not the best representation for this specific data set, and that there may have been some errors in the data collection. This is because theoretically, if the velocity squared is equal to 0, the lift force should too be 0. Therefore, this graph does support the hypothesis that the lift force will increase as the velocity squared increases.

The gradient of the curve 0.0123 represents the constant k in the hypothesis equation, which is also equal to the product of the coefficient of lift, the surface area of the wing and the density of air.

Coefficient of lift:

With the data that was collected, I realized that it was actually possible to dive one step further, and calculate the coefficient of lift, since the constant k was actually equal to the product of the coefficient of lift, the surface area of the wing and the density of air. First however, it was important to show how the density of air was calculated.

Finding density of air:

However, to find the coefficient of lift, we must first find the density of air. To find the value of the density of air at any given location, the air temperature (T), relative humidity (RH) and air pressure (p) were needed. (Air Density Calculator, omnicalculator, 2022).

The steps below are the derivation of the density of air formula, taken from (Air Density Calculator, omnicalculator, 2022), which I used to calculate the formula of the air density.

To calculate the water vapour pressure, it is first needed to calculate the saturation vapour pressure at the given temperature. As such, the formula below was used, where p_1 refers to the saturation vapour pressure, and T refers to the temperature in degrees Celsius.

$$p_1 = 6.1078 * 10^{\left(\frac{7.5 * T}{(T + 237.3)}\right)}$$

Following this, since saturation vapour pressure refers to the pressure when the humidity is 100%, this value p_1 has to be multiplied by the relative humidity (RH) of the air.

$$pv = p_1 * RH$$

To find the pressure of dry air, a simple subtraction of the total air pressure (p) by the water vapour pressure can be used.

$$pd = p - pv$$

By combining the details from the above equations, a new equation can be derived to find the density of air, as shown below.

$$\rho = (pd / (Rd * T)) + (pv / (Rv * T))$$

Where ρ is the density of air, pd refers to the pressure of dry air in Pa, pv refers to water vapour pressure in Pa, T is the temperature in Kelvins, Rd is the specific gas constant for dry air equal to 287.058 J/(kg·K), and Rv is the specific gas constant for water vapor equal to 461.495 J/(kg·K).

To calculate the density of air, the pressure, temperature and relative humidity had to be found. These three values were found using the Kestrel 3000 windspeed meter.

Table 6 (Table showing raw data from Kestrel 3000 windspeed meter)

Relative Humidity (%)	Temperature (°C)	Air Pressure (hPa)
76.5	25.4	1005.6

By keying these results into the equation, it is hence possible to find that the density of air is 1.16kg/m³.

Since I knew that the gradient of the curve (0.0123) is the product of the coefficient of lift, the surface area of the wing and the density of air, I could hence calculate the coefficient of lift in this wing. Since the surface area of the wing is to be in m^2 , which means that it has to be converted from cm^2 to m^2 . Since we know that the Surface area of the wing is $0.00589m^2$ (from raw data) and the density of air is $1.16kg/m^3$, we can calculate the coefficient of lift as shown below.

$$\begin{aligned} \text{Gradient} &= Cl * \rho * A \\ 0.0123 &= Cl * 1.16 * 0.00589 \\ Cl &= \frac{0.0123}{1.16 * 0.00589} \\ Cl &= 1.80 \end{aligned}$$

Conclusion

In conclusion, the hypothesis can be supported by the data. From the graph, all of the six error bars were seen to pass through the line of best fit. This shows that the data points were indeed useful for showing the relationship between the velocity squared and the lift force generated. From this, it can be observed that the data is linearly related but not directly proportional. (Chris Hamper, 2014) However, since the line of best fit did not really pass through the origin, there may have been a systematic error in mass. For example, if the balance had not been zeroed properly and instead of displaying zero with no mass on the pan it read 0.5 g then each mass measurement would be 0.5 g too big. (Chris Hamper, 2014)

The graph and data however, does support the research question, which investigates the effects of changing wind speeds on the lift force generated. The lift force generated was found to be linearly related to the windspeed squared. The gradient of this was then used along with some other primary data to find the coefficient of lift, which was found in this case to be 1.80. Since the coefficient of lift changes in every plane and every type of aerofoil, there was no literature value to compare this to. However, on doing some research, I found that the coefficient of lift for most simple aerofoils have a maximum of approximately 1.7-1.8. (Airfoil Theory, 2003) This meant that the values found in the investigation were within limits, and hence the coefficient of lift of this aerofoil was within limits.

Evaluation

Although the data gathered does show a strong similarity with the theoretical values and relationship that was being tested, errors were still present that may have caused the data to be inaccurate. Some of these errors were unavoidable, due to the inaccuracies and limitations of the instruments that were used in a school lab.

Strengths and weaknesses (experimental design)

Pros:

- A string tied the mass to the aerofoil, ensuring that the aerofoil was always on the weighing scale and did not float independently if the windspeeds increased.
- The wind tunnel ensured that the aerofoil was not influenced by other possible sources of moving air currents, isolating the system.
- A second string was passed through the aerofoil, which ensured it stayed upright at all times, and did not change in angle of attack throughout the experiment

Cons:

- The experiment was very time consuming and difficult to carry out.
- The aerofoil was fairly flimsy, and so at higher windspeeds it tended to vibrate a fair bit, and sometimes even rotate about the string.
- The fan was a bit inconsistent in windspeeds, and so would take time to stabilize.

Strengths and weaknesses (Procedure)

Pros:

- The use of 3 readings reduced the likelihood of random and human error, and this improved the reliability of the results.
- The graphs plotted on logger pro were very easy to analyze and this allowed for easy observation of various patterns and correlations
- Surrounding fans were turned off which meant that even without the wind tunnel the surrounding air would still be fairly still.

Cons:

- The fan had no preset settings, and so had to be adjusted before every trial.
- There was no fixed position to measure the windspeed, and so by merely putting the meter in front of the aerofoil, it may have measured a windspeed slightly different from what the aerofoil was actually receiving.
- The weight was constantly changing, fluctuating slightly, and so the instances where the mass stabilized may not have been the best representation of the actual weight at that windspeed.

Possible improvements to the experiment:

Although the experiment did provide us with most of the data that was needed to demonstrate the trend, hypothesis, and to answer the research question, there are a few changes that could be made in both the setup and the methodology. For example, changing the material of the aerofoil would have made it much more stable, decreasing the flimsiness of the aerofoil and making the readings more stable. Furthermore, the setup could be improved by merely supporting the aerofoil on three stilts glued to a cardboard base instead of a string and a straw, as this would have prevented the rotation of the aerofoil, and decreased the amount of inaccuracies in data collected.

The procedure could have been modified such that the windspeed meter was placed in a fixed position that would be swapped for the aerofoil when the windspeeds stabilized, instead of placing the windspeed meter in front of the aerofoil. To help correct the graph however, an alternative method could be used, where a camera with slow motion is set up, and the windspeed meter is placed next to the aerofoil. When the fan is turned on, the camera can record the exact masses when the windspeeds are at the exact timings, and this will reduce the fluctuations in the masses, increasing reliability of data, and it will probably be reflected in the graph, where the data will probably be less inaccurate, and fit the trend line even better.

Possible extension questions:

A possible extension question would be to research and find how the coefficient of lift may vary in the angle of attack of a wing, as the same experiment could be carried out, but with the angle of attack being the independent variable instead of the windspeed.

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